Exploring Meaning Beyond the Axes: A Comparative Study of PCA Interpretability in High-Dimensional Representation Spaces

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ABSTRACT

Principal Component Analysis (PCA) is a standard tool for dimensionality reduction, yet its interpretability varies dramatically across domains. This work examines why this happens by comparing PCA's behavior across two fundamentally different representation forms: learned word embeddings and feature-based tabular data. Using pretrained GloVe embeddings, PCA successfully uncovers a dominant semantic direction derived from definitional word pairs, producing a clear separation in the embedding space and retaining much of the discriminative power of the original 300-dimensional vectors. In contrast, applying the same procedure to the Wine Quality dataset results in principal components that are abstract mixtures of heterogeneous chemical features, offering no coherent interpretation despite capturing statistical variance. Together, these findings reveal that PCA outputs interpretable components only when the data contains strong linear structure, otherwise, it merely compresses variance without revealing meaningful latent factors.

1. INTRODUCTION

High-dimensional data representations have become integral to modern data analysis and machine learning. From genomic sequences with thousands of gene expression levels to customer behavior captured through hundreds of interaction metrics, many datasets often encode information across far more dimensions than humans can intuitively reason about. This phenomenon reflects what is broadly known as the *curse of dimensionality*, describing how computational and analytical complexity grows rapidly as dimensionality increases. Although these representations capture complex patterns and support highly effective predictive systems, they also introduce a fundamental challenge: how can one interpret structure when it is distributed across several dimensions that may not correspond to intuitive, human-intelligible concepts? In both engineered feature spaces and learned representation spaces, individual dimensions lack direct semantic meaning, creating a disconnect between representational capacity and human understanding.

Principal Component Analysis (PCA) is one of the most widely used methods for analyzing such spaces. As a dimensionality-reduction technique, PCA identifies orthogonal directions of maximum variance in the data and projects it into a lower-dimensional representation. It is commonly used for visualization, compression, noise reduction, and exploratory analysis. However, PCA does not guarantee interpretability of its output. In some contexts, the principal components capture meaningful structure encoded in the data. In other cases, PCA produces abstract mixtures of features that obscure meaning rather than clarifying it. Understanding this dual behavior is essential for determining when PCA is useful as an interpretive mechanism, and when it is not.

This research investigates this issue by comparing PCA's behavior across two fundamentally different types of high-dimensional representation spaces: **word embeddings** and **tabular data**. These data types differ strongly in how they encode semantic or structural information. Word embeddings are learned vector representations in which linguistic regularities are embedded in several linear directions, making their individual dimensions hard to explain, but their geometric relations meaningful. PCA may therefore recover semantic axes from the global structure. In contrast, tabular datasets consist of explicit and interpretable features such as age, income, or biological measurements. Applying PCA to such data typically mixes these meaningful dimensions into abstract combinations, reducing interpretability even if variances is preserved.

To formalize our goal, the main research question guiding this study can be defined as:

RQ: *How does PCA influence interpretability across different forms of high-dimensional data, and what types of latent information can it successfully recover?*

The central hypothesis of this work is that components resulted through PCA reveal meaning in abstract, learned data representation forms with underlying linear semantic directions. When the data lacks this kind of global structure, as in typical tabular feature sets, PCA instead blends distinct variables into composite components that obscure meaning and are difficult to interpret.

To provide a comprehensive analysis, the report is structured as follows. First, prior work is reviewed to situate PCA within the broader discussions of interpretability in high-dimensional spaces. Next, the datasets and experimental methodology are described, followed by an analysis of the obtained results using PCA applied to both selected data domains. The final part of the report discusses the implications of the comparative findings and outlines limitations and possible directions for future research.

2. RELATED WORK

Research on high-dimensional data analysis spans several complementary areas, including dimensionality-reduction techniques, the structure of learned representation spaces, and methods for understanding how meaning or variation is encoded in such spaces. This section reviews prior work relevant for the themes that are later explored in the conducted experiments, focusing on two main concepts: research on PCA and the interpretability of its components, and studies examining the selected data representation forms.

2.1 Principal Component Analysis: Foundations and Interpretability

Principal Component Analysis (PCA) represents a classical linear dimensionality-reduction technique widely used in statistics, machine learning, and exploratory data analysis [1]. Its objective is to transform a high-dimensional dataset into a new coordinate system whose axes, known as **Principal Components** (PCs), successively maximize the variance of the projected data. This means that the identified orthogonal directions capture dominant patterns in decreasing order of importance, filtering noise and reducing the representation space to analyze. Because of its simplicity and effectiveness, PCA remains a fundamental tool for understanding high-dimensional structure.

Formally, let $\tilde{X} \in R^{n \times d}$ denote the mean-centered matrix associated to a dataset with n samples and d features. PCA begins by computing the sample covariance matrix $\Sigma = \frac{1}{n-1} \widetilde{X^{\top}} \tilde{X}$, and performs its eigen-decomposition $\Sigma = V \Lambda V^{\top}$, where $V = [v_1, \dots, v_d]$ contains orthonormal eigenvectors and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$ contains eigenvalues sorted in descending order. The eigenvectors correspond to directions that diagonalize the covariance matrix, meaning they identify uncorrelated axes ordered by the amount of variance the explain. Each eigenvalue λ_k quantifies the variance explained by the corresponding principal component, while each eigenvector v_k defines a principal direction. Projecting a centered data point $\tilde{x}$ onto the k-th principal component yields $z_k = v_k^{\top} \tilde{x}$, which shows that each principal component is a linear combination of the original features: $\mathrm{PC_k} = \sum_{j=1}^{d} v_{kj} \widetilde{x_j}$. The loadings v_{kj} indicate how much each original feature contributes to the component. Dimensionality reduction is achieved by retaining only the first $k \ll d$ components, producing the transformed representation $Z = \tilde{X} V_k$.

Geometrically, PCA performs an orthogonal rotation of the coordinate system such that the first axis aligns with the direction of maximum variance in the data, the second axis aligns with the maximum remaining variance while being orthogonal to the first, and subsequent components follow the same pattern. In a typical two-dimensional representation, this often corresponds to an elongated cloud of points whose major axis coincides with PC1 and whose minor perpendicular axis corresponds to PC2. The eigenvalues associated with each component quantify the amount of variance captured, and arranging them in a *scree plot* provides a visual summary of how effectively the reduced set of components preserves the total variability of the data.

While PCA provides a powerful geometric summary of a dataset, its interpretability if often reduced. Since each principal component represents a mixture of potentially all original variables, the resulting axes do not necessarily correspond to meaningful concepts. PCA focuses on maximizing statistical variance rather than semantic coherence, meaning that the most dominant direction in the data may capture correlations, noise patterns, or broad-scale variability rather than interpretable latent factors. While this is a helpful transformation for both visualizing the original high-dimensional data and for training predictive machine learning models, which make better use of less but more relevant features, when it comes to attempting to identify certain trends that explain the data distribution, principal components may prove difficult to understand or associate with human-intelligible terms. Overall, this highlights a core challenge of PCA: maximizing variance does not inherently mean obtaining meaningful interpretation, further motivating a deeper investigation into the method's interpretive capability.

2.2 Semantic Structure in Word Embedding Spaces

In contrast to standard tabular datasets whose dimensions correspond to explicit, human-defined features, learned embedding spaces organize information in fundamentally different ways. Word embedding models such as Word2Vec [2] and GloVe [3] generate dense vector representations by exploiting the distributional hypothesis, assigning similar vectors to words that occur in similar linguistic contexts. As a result, the meaning of a word is not contained in a single dimension but is distributed across several numerical coordinates, each capturing subtle patterns of co-occurrence and contextual usage. A well-documented consequence of this distributed structure is that many linguistic relations correspond to *approximately linear directions* in the high-dimensional embedding spaces, highlighted through vector analogies [2]. For example, the well-known analogy pattern $king - man + woman \approx queen$ illustrates how certain semantic contrasts (e.g., gender, polarity, tense) tend to be encoded in the geometry of the latent space, as consistent linear vector offsets. Shifting a word vector along such a direction applies the corresponding semantic transformation. This property directly motivates the use of a dimensionality reduction tool as PCA, in order to recover concept directions effectively.

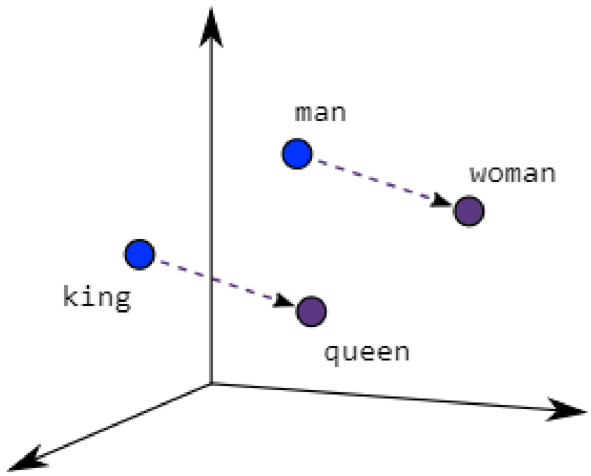

Figure 2.1. Illustration of vector offsets in the word embedding space of Word2Vec model [2] , showing how linear directions encode semantic transformations.

One influential line of work that leverages this property for a similar purpose is the study of *bias subspaces* in word embeddings [4]. Bolukbasi et al. demonstrated that embedding models encode not only semantic relations, but also stereotypical associations inherited from human text, such as gender bias in words describing occupations. By using *concept-definitional word pairs*, they demonstrated that PCA applied to the corresponding contrast vectors reliably captures a semantic direction as the dominant axis of variation, which is then further eliminated to mitigate the captured biases.

This insight directly motivates the present research and provides a conceptual foundation for the experiments, suggesting that PCA's interpretability depends critically on the structure of the underlying representations to which it is applied. Consequently, a comparative approach is adopted, contrasting abstract embedding spaces that encode linear relations, and tabular datasets consisting of human-defined features.

3. METHODOLOGY

This section has the goal of outlining the experimental framework adopted to investigate how the interpretability of PCA depends on the nature of the high-dimensional representations to which it is applied. This study follows a comparative design involving two fundamentally different data modalities. The first experiment analyzes word embedding vectors and explores the hypothesis that semantic relations encoded geometrically in their representation space can be successfully revealed using PCA. The second experiment uses PCA on a classical tabular dataset, specifically the Wine Quality dataset, whose heterogenous features do not form comparable geometric structure, and therefore their linear combinations may not be assignable to human-intelligible concepts.

By applying the same dimensionality reduction method across both modalities and evaluating the resulting components, projections, and visualizations, we aim to highlight the conditions under which

PCA can also be considered an interpretability tool for the data and the contexts in which it does not bring an advantage.

3.1 PCA on Word Embedding Spaces

Word embeddings represent a family of learned, high-dimensional vector spaces designed to encode semantic and syntactic properties of words. They appeared as a consequence of emerging research interest in the field of Natural Language Processing (NLP), more specifically for addressing a fundamental need: representing language in a continuous numerical format that preserves contextual information, similar to how the human brain does. Unlike simple one-hot encodings, which assign each word an orthogonal basis vector incapable of modeling meaning, embedding models learn dense vectors in which semantic similarity corresponds to geometric proximity. In these vector spaces, meaning is spread across many dimensions, because the model learns patterns from how words appear together in text.

Out of the existing models in this field, in this experiment, we select the GloVe model [3], which exploits global co-occurrence statistics for learning word embeddings. It constructs a large, 300-dimensional, *word-word co-occurrence matrix* X, where X_{ij} counts how often word j appears in the context of word i. The model's key insight is that semantic information is reflected in the ratios of co-occurrences rather than raw counts. GloVe therefore learns word vectors w_i and context vectors $\widetilde{w_j}$ such that their dot product approximates $log\, X_{ij}$, optimized through a weighted least-squares objective. This formulation produces embeddings that prove to capture semantic relations manifested as linear vector offsets, enabling geometric transformations that can help explain the obtained representation space.

Inspired by the linearly interpretable structure of such an embedding space and the experiments conducted for debiasing word embeddings from stereotypes present in our language [4], we choose to formulate a similar methodology, but with the ultimate goal of analyzing the interpretability of PCA. More specifically, we choose to explore human-defined concepts that can be expressed as contrasts and therefore, correspond to linear directions, and we utilize PCA in order to visualize and evaluate a dataset in a projected representation space corresponding to those concepts. We assess how the obtained components model the data points in such a way so that the searched concept becomes visible and explainable.

The conceptual focus of this experiment is **gender**, a semantic dimension intensively studied in prior work and shown to be represented as a consistent linear direction in embedding spaces. Since it is a concept that humans naturally perceive as a binary contrast, PCA is expected to recover a direction that can separate male and female associated terms. For this purpose, we use the gender-labeled word lists dataset introduced in the GN-GloVe framework [5], which were curated from 2017 English Wikipedia dump to study an mitigate gender bias in distributional embeddings. The dataset contains a diverse set of occupation names, role nouns, and general vocabulary items annotated as male-coded or female-coded base on conventional human understanding. These collections of 222 words per gender are then embedded using the 300-dimensional GloVe model, providing a high-dimensional representation space suitable for analyzing how PCA captures this semantic distinction.

Following Bolukbasi et. al. [4], we begin by constructing a concept definitional matrix using definitional word pairs that differ primarily along the gender dimension (e.g., man-woman, king-queen, boy-girl, brother-sister). These pairs satisfy the following criteria:

- **Semantic comparability**: Both words must belong to the same lexical category ($noun \leftrightarrow noun$).
- **Primary variation along gender concept:** They must differ primarily on the chosen concept, other contextual differences must be minimal.
- **Sufficient frequency**: They must be common enough in the language to have stable embeddings and enough contextual usage in the chosen word dataset.

For a definitional pair (m_i, f_i), we construct a *centered difference vector*, defined as $d_i = w_{m_i} - \frac{w_{m_i}+w_{f_i}}{2}$ and $d_i' = w_{f_i} - \frac{w_{m_i}+w_{f_i}}{2}$. Centering removes the shared semantic content of the two words and isolates only the contrast that varies along the gender dimension. Stacking these difference vectors results in the matrix $D = [d_1^\top, d_1', d_2^\top, \cdots, d_k^\top] \in R^{2k \times n}$, where k is the number of definitional pairs and $n = 300$ is the embedding dimensionality. Each row represents a semantic contrast, and each column corresponds to an embedding dimension. Therefore, difference vectors can be treated as *observations*, and embedding dimensions as *features*. In this experiment, we arbitrarily choose the definitional pairs for specifying the gender concept as follows: $(\text{woman}, \text{man}), (\text{she}, \text{he}), (\text{girl}, \text{boy})$. They have been chosen because they differ primarily along gender while minimizing unrelated semantic differences.

PCA is then applied to the centered matrix D to identify orthogonal directions that successively maximize variance among the definitional difference vectors. Let $(v_1, v_2, \cdots, v_n)$ denote the principal components obtained from PCA. Each principal component $v_j \in R^n$ represents a linear combination of embedding dimensions. The explained variance ratio for each component is computed as: $\mathrm{EVR(j)} = \frac{\lambda_j}{\sum_{i=1}^{n} \lambda_i}$, where λ_j is the eigenvalue associated with component v_j. This measure indicated the proportion of variation in D accounted for by each principal component, and is used to analyze whether the method is able to identify the gender contrast as the main difference between the data points.

To inspect how the discovered principal components relate to the broader gendered vocabulary, each word embedding w is projected into the PCA basis. The projection onto the first principal component (PC1) is computed as $\frac{v_1^\top}{|v_1^\top|}\mathrm{w}$ which we denote as $GenderScore(w)$. This scalar quantifies how strongly a word aligns with the gender direction identified from definitional pairs. For visualization, 2D coordinates are obtained by projecting words onto the first two PCA components. These projected coordinates can be used for visual inspection of separation patterns between male-coded and female-coded words, which should reveal if the generated semantic axis is indeed capable of capturing the meaning of the investigated concept.

These procedures collectively assess the extent to which PCA uncovers interpretable conceptual structure in high-dimensional learned embedding spaces.

3.2 PCA on Tabular Data Representations

Tabular datasets differ fundamentally from learned embedding spaces. Their dimensions correspond to explicit, human-defined features than latent geometric factors created to encode patterns. They represent measurable quantities, such as physical, chemical, biological, or statistical attributes. Because such variables do not necessarily form coherent latent directions and may vary on incompatible scales, PCA often produces principal components that combine heterogenous features, making interpretability more challenging. This second experiment applied PCA to a classical tabular dataset to assess how the method behaves when applied to structured, feature-based representations of data, that may not inherently encode semantic relations.

The Wine Quality dataset [6] is selected as a representative high-dimensional tabular dataset frequently used in data analysis and predictive modeling tasks. It consists of physiochemical measurements of red and white wines, including attributes such as fixed acidity, residual sugar, pH, alcohol content, chlorides, sulphates, and density, summarized in Figure [3.1]. Each observation corresponds to one wine sample, and each feature represents a measurable property recorded through laboratory analysis. This dataset provides a suitable contrast to word embeddings. Although each dimension is individually interpretable, the dataset as a whole does not possess an inherent semantic structure beyond the statistical correlations among features.

	count	mean	std	min	25%	50%	75%	max
fixed acidity	1143.0	8.311	1.748	4.6	7.1	7.9	9.1	15.9
volatile acidity	1143.0	0.531	0.18	0.12	0.392	0.52	0.64	1.58
citric acid	1143.0	0.268	0.197	0.0	0.09	0.25	0.42	1.0
residual sugar	1143.0	2.532	1.356	0.9	1.9	2.2	2.6	15.5
chlorides	1143.0	0.087	0.047	0.012	0.07	0.079	0.09	0.611
free sulfur dioxide	1143.0	15.615	10.25	1.0	7.0	13.0	21.0	68.0
total sulfur dioxide	1143.0	45.915	32.782	6.0	21.0	37.0	61.0	289.0
density	1143.0	0.997	0.002	0.99	0.996	0.997	0.998	1.004
pH	1143.0	3.311	0.157	2.74	3.205	3.31	3.4	4.01
sulphates	1143.0	0.658	0.17	0.33	0.55	0.62	0.73	2.0
alcohol	1143.0	10.442	1.082	8.4	9.5	10.2	11.1	14.9

Figure 3.1. Statistical summary of the Wine Quality Prediction dataset features, including mean, standard deviation, and quartiles.

PCA is conventionally applied to this dataset for several reasons. First, many of the variables are moderately correlated (e.g., density with sugar content, acidity with pH), which motivates dimensionality reduction as a means of mitigating multicollinearity before applying machine learning models such as linear regression, SVMs, or neural networks. Second, PCA enables visualization of the data in a lower-dimensional space, facilitating exploratory analysis, outlier detection, and preliminary understanding of variance structure. Ultimately, PCA can help compress redundant information and improve numerical stability without requiring domain-specific feature engineering, yet these inherent advantages may directly affect the interpretability factor of the reduced resulted latent representation space.

A subset related to red variants of wine is employed for the experiment, containing $n = 1143$ samples and $d = 11$ features. Since PCA is sensitive to scale differences, all features are standardized before analysis. Let the original dataset be represented as $X \in R^{n\times d}$, where rows correspond to wine samples and columns to physiochemical attributes. Standardization is performed feature-wise: $\widetilde{X_{ij}} = \frac{X_{ij}-\mu_j}{\sigma_j}$, with μ_j and σ_j denoting the mean and standard deviation of feature j. This transformation ensures that all variables contribute equally to the PCA computation, preventing high-variance features (e.g., alcohol percentage) from dominating the principal components. No further feature engineering is performed, as the goal of this experiment is not predictive modeling, but rather evaluating PCA's interpretability on raw feature-based data.

PCA is applied to the centered, standardized matrix $\tilde{X}$ to obtain principal components that capture directions of maximal variance in the feature space. PCA proceeds by computing the sample covariance matrix: $\Sigma = \frac{1}{n-1}\widetilde{X^{\top}}\tilde{X}$, and performing eigen-decomposition: $\Sigma = V\Lambda V^{\top}$. Unlike the embedding experiment, where PCA is applied to difference vectors, PCA here operates directly on raw feature matrix. Thus, the principal components represent linear combinations of physiochemical measurements rather than conceptual contrasts. Similarly, we visualize the structure captured by PCA by projecting the wine samples onto the first two principal components, looking for patterns or underlying data trends. Additionally, the component loading vectors v_1, v_2 will be examined to analyze how the original features contribute to each principal direction. Because the features are semantically interpretable individually, the loadings provide insight into how PCA mixes different physical measurements when constructing components, as our goal is to evaluate how the computed latent directions can be associated in any way with human-intelligible concepts which address the relations between the original features.

4. RESULTS AND DISCUSSION

The results of the two experimental settings of PCA applied to different data forms are presented and analyzed in this section. The goal is to evaluate how effective PCA uncovers interpretable structure across fundamentally different high-dimensional representation spaces. By comparing variance structure, component loadings, geometric projections, and classification behavior across both domains, we highlight the conditions under which PCA can capture explainable concepts, but also in which scenarios variance maximization fails to align with semantic meaning. The discussion that follows integrates both visual and quantitative findings to assess the role of PCA as an interpretability tool rather than solely a dimensionality reduction method.

4.1 Where PCA Succeeds: Interpreting Semantic Axes in Learned Embeddings

Based on the principal components obtained by applying PCA to the constructed definitional matrix for the gender concept, we report the bar plot of **Explained Variance Ratio** (EVR) of each PC in Figure [4.1], in order to quantify the total variance in the definitional vectors captured by each principal direction. The first component explains **75.7%** of the variance, followed by much smaller contributions from a reduced number of subsequent components. For example, PC2 explains **12.5%** and PC3 only **11.8%**. This sharp dominance of PC1 indicates that there is a single, strongly aligned direction across all definitional difference vectors, consistent with the hypothesis that gender can be represented as a linear semantic axis. This one-dimensional structure does not appear in arbitrary or unrelated concepts (e.g., age), confirming that PCA is correctly isolating a meaningful factor.

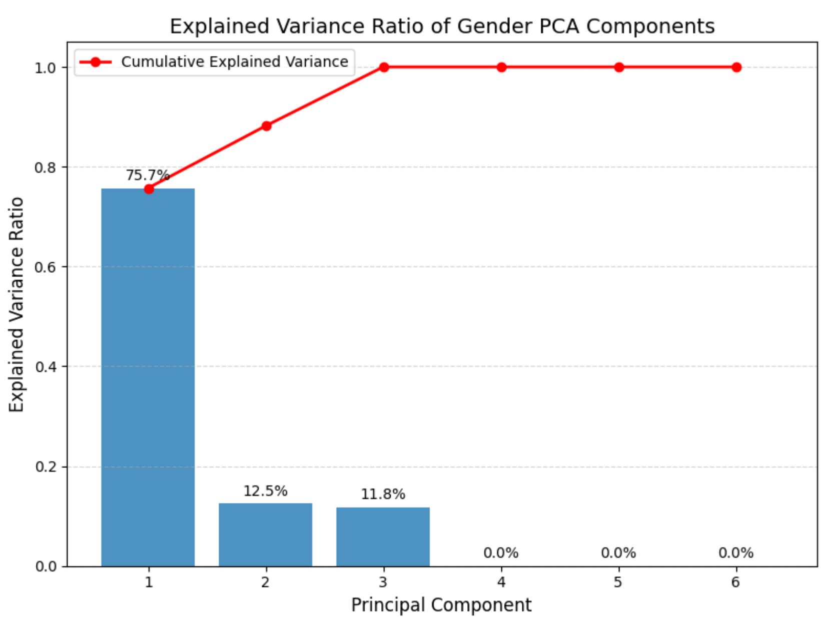

Figure 4.1. Explained variance ratio of the principal components derived from the gender definitional matrix.

From the obtained PCs, we select the first two components and we project all gender-labeled words of the dataset onto them for a clearer 2-D visualization of the gender subspace, seen in Figure [4.2]. The resulting plot shows a linear separation between male-coded and female-coded words along the axis corresponding to the first principal component. Male-associated words cluster predominantly on the positive side of the axis, while female-associated words cluster on the opposite side. Since most of the words In the dataset also represent occupation names or other concepts that contain gender information, or have gender stereotypes encoded, the separation is not perfect, as there may exist other associations between the data points besides gender. However, the goal of obtaining a subspace which can characterize the data strictly from the gender perspective is achieved through this experiment, demonstrating that the chosen defining semantic factor is encoded as dominant.

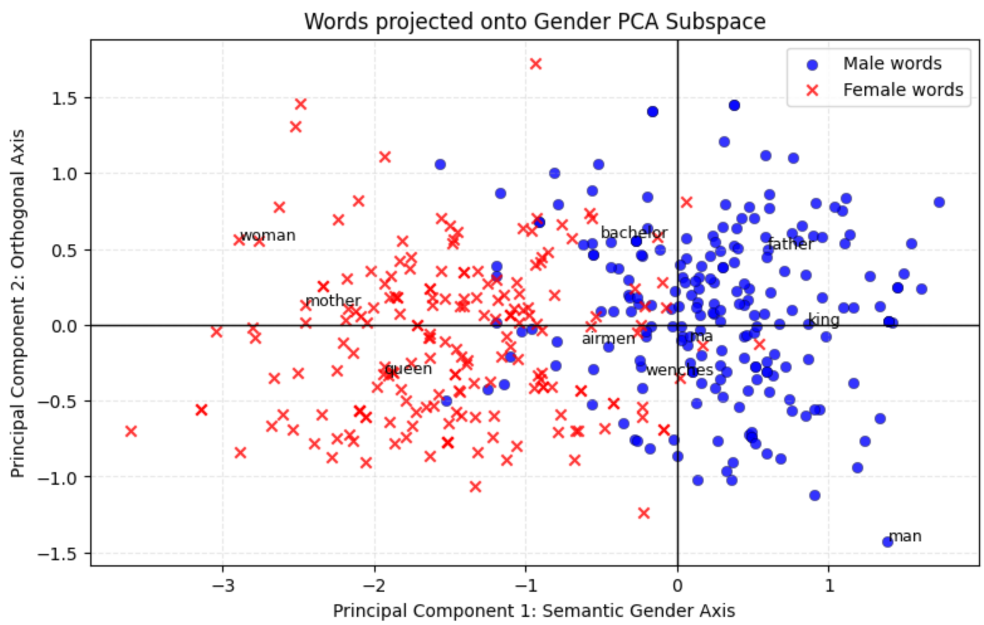

Figure 4.2. 2-D PCA projection of gender-labeled words. PC1 aligns with the gender contrast, while PC2 forms an orthogonal residual axis.

Additionally, Figure [4.3] shows the PC1 loadings across all 300 embedding dimensions. The loadings v_{1j} indicate the contribution of each embedding dimension to the desired gender direction: $\mathrm{PC1} = \sum_{j=1}^{300} v_{1j} x_j$. The plot reveals that no single embedding dimensions encodes gender. Instead, gender is a distributed semantic feature spread across many dimensions of the original data. Positive and negative peaks correspond to dimensions that consistently increase or decrease for male-coded vs. female-coded words. This distributed pattern justifies the need for dimensionality reduction. PCA aggregates a dispersed semantic signal into a single interpretable conceptual axis.

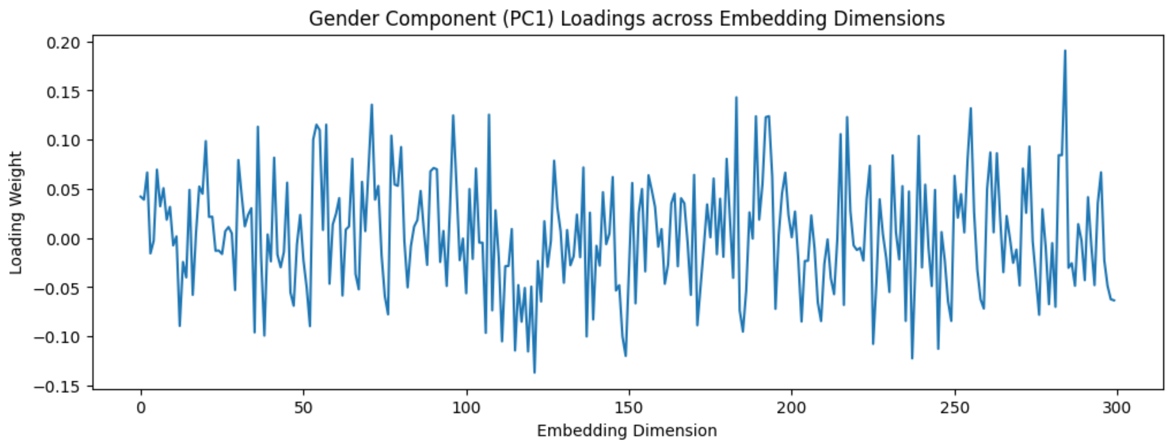

Figure 4.3. Loadings of the first principal component across the 300 embedding dimensions, showing that the gender signal is distributed rather than concentrated in any single dimension.

Using the normalized gender direction $g = \frac{v_1}{|v_1|}$, each word is assigned to a gender score $GenderScore(w) = g^\top w$, based on a dot product between that direction and the word embedding. It is a projection that measures the signed strength with which a word aligns with the discovered semantic direction. The score distribution in Figure [4.4] shows the histograms for male-coded and female-coded words. The distributions show a reduced overlap, similar to what was observed in the 2-D projection plot, confirming that PC1 serves as a discriminative semantic direction capable of distinguishing gender associations in the vocabulary.

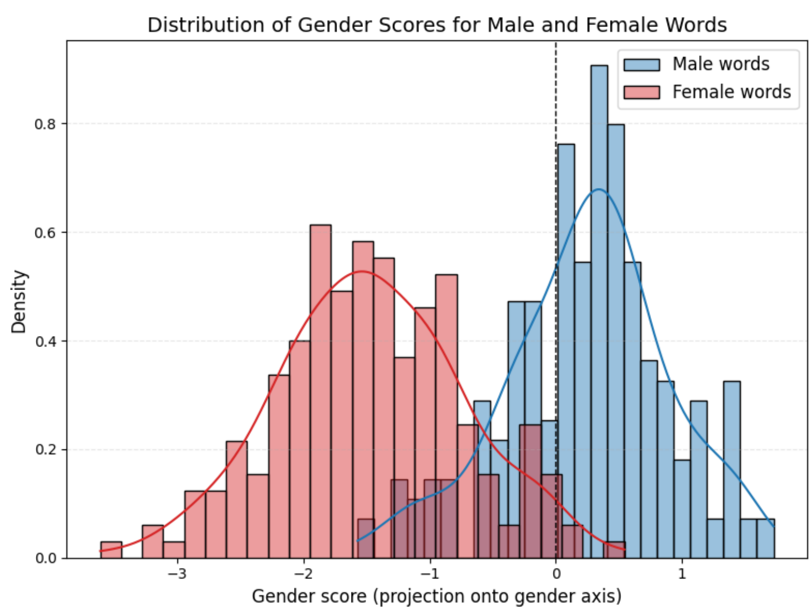

Figure 4.4. Distribution of gender scores for male and female associated words.

In order to assess the quality of the PCA-derived gender direction, we evaluate the projection onto PC1 based on the proposed gender score. Although PCA is not a classification algorithm, its first principal component can be interpreted as a one-dimensional semantic axis, and the signed projection $g^\top w$ serves as a measure of how strongly each word aligns with this direction. Treating these scores as classifier outputs, we compute the Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) as seen in Figure [4.5], which quantifies how well PC1 arranges male vs. female words. The resulting AUC of **95.7%** indicates that a single principal component provides a strong separation in the subspace. For comparison, a logistic regression classifier trained on the full 300-D embeddings achieves an AUC of **96.8%**, only marginally higher, supporting how the gender-relevant information present in the full embedding vectors can be captured by the unsupervised PCA direction.

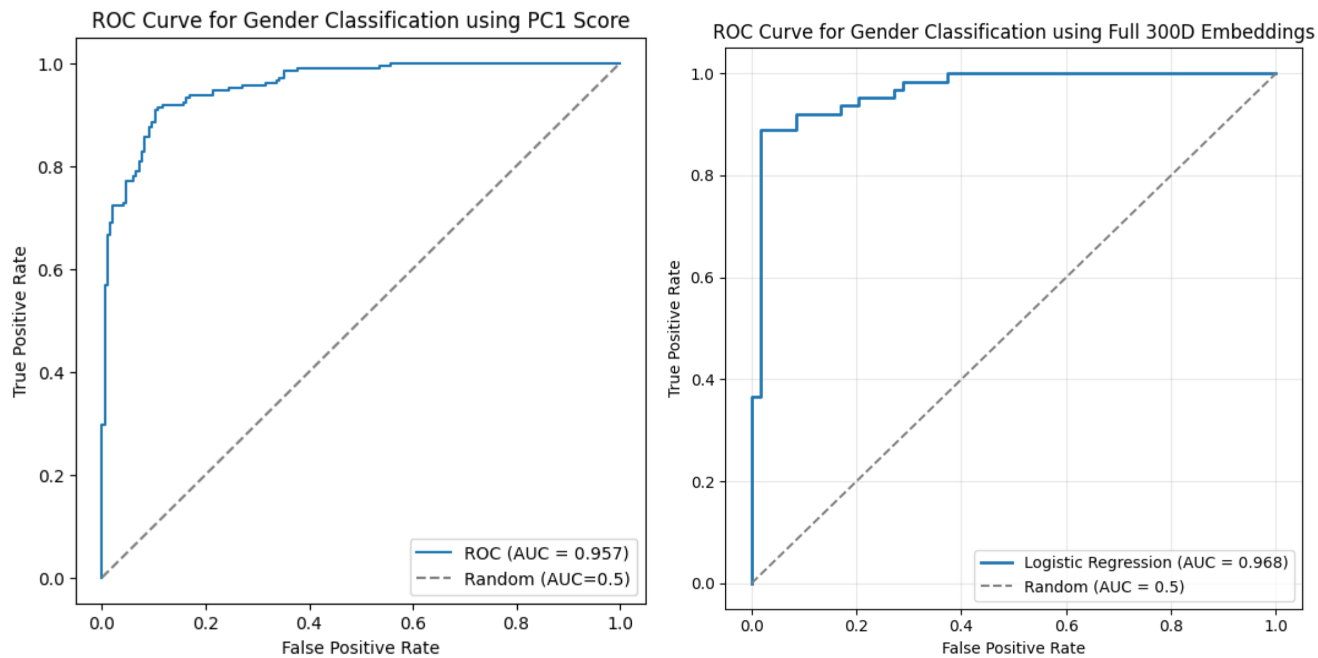

Figure 4.5. Comparison of ROC curves for gender prediction using PCA gender axis (left) and a classifier trained on the full embedding space (right).

Despite the strong interpretability of the recovered gender axis, this approach has several limitations. First, the PCA direction is sensitive to the choice of definitional word pairs: adding pairs such as mother-father or sister-brother often reduces the clarity of the gender principal component because these words encode additional semantic relations (e.g., family roles) that introduce other linear patterns. More fundamentally, this technique applies only to concepts that correspond to approximately linear contrasts in the embedding space, as it is the case of gender. However, for many other more complex semantic properties, such as sentiments, PCA fails to isolate a meaningful axis, and the resulting components mix multiple unrelated factors, limiting interpretability.

4.2 Where PCA Fails: Revealing Interpretable Structure in Feature-Based Datasets

A similar analysis pattern and performance evaluation process is conducted for the second experiment, applying PCA to the original dataset with the 11 physiochemical measurement features, ignoring the quality feature representing the result of a correct prediction, and also the ID column. Figure [4.6] presents the explained variance ratio of the principal components extracted from the standardized feature matrix. In contrast to the embedding experiment, no single component accounts for a dominant proportion of the variance. The first principal component explains only **28.7%** of the variance, with subsequent components contributing distributed amounts ranging from less than 1% to 17%. The absence of a sharp drop-off indicates that variance in this dataset is spread across many unaligned dimensions. This pattern suggests that there is no strong, coherent latent direction in the Wine Quality feature space. Instead, the dataset contains multiple weak correlations among heterogenous chemical properties, none of which emerge as a primary axis of variation, or in a combination with others, remain just an abstract mixture of features. This is consistent with the expectation that PCA has limited interpretability in such settings.

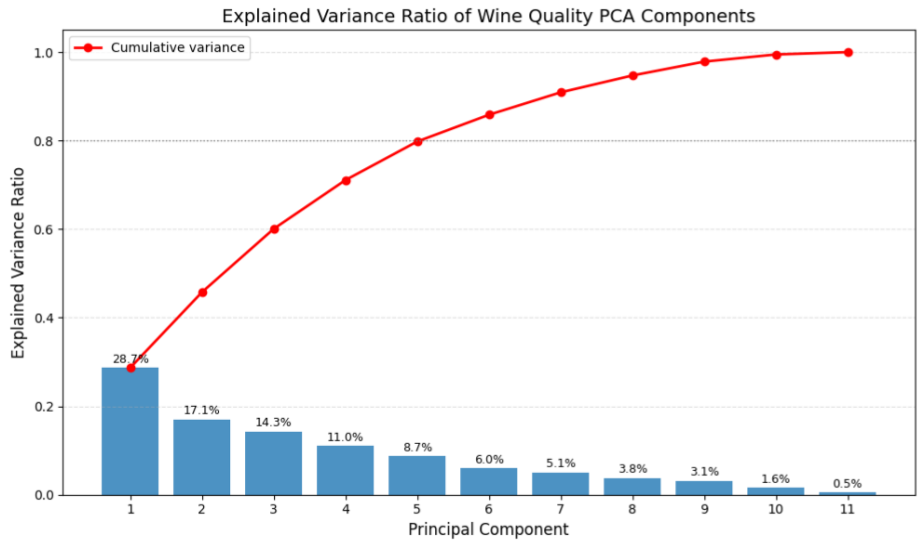

Figure 4.6. Explained variance ratio of PCA components for the Wine Quality dataset. No single component dominates, indicating that variance is spread across many unaligned dimensions.

The 2D projection of the standardized samples on the first two PCs from Figure [4.7] shows a compact spread of data points with substantial overlap between sines of different quality levels. Unlike the word embedding projection, which showed a clear separation in the data, the Wine Quality plot reveals no meaningful grouping of samples or structure related to the target variable. This suggests that the dominant sources of variance in the dataset do not correspond to interpretable factors. The PCA projection therefore provides little insight into the nature of wine quality.

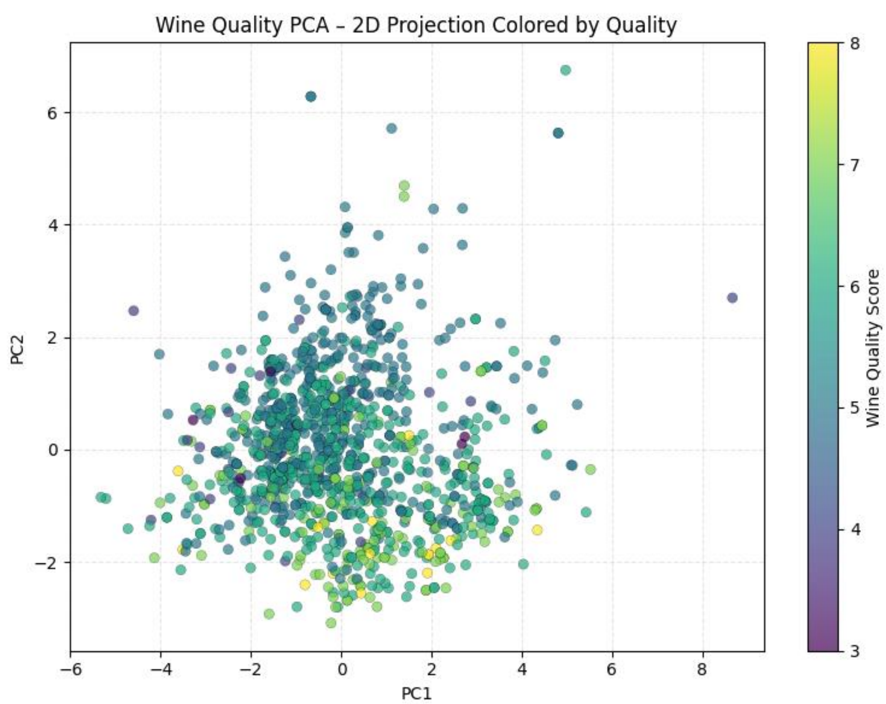

Figure 4.7. 2-D PCA projection of wine samples, colored by quality score. No clear clustering or structure aligned with the wine quality labels.

Figure [4.8] displays the loadings for PC1 and PC2, corresponding to the contribution of each original feature to the components. Both components assign substantial weight to several different features with no coherent interpretation, while others contribute only marginally. This pattern shows that the PCs do not correspond to meaningful dimensions such as acidity, sweetness or aroma for example. Instead, they reflect arbitrary correlations within the data, highlighting that PCA cannot extract interpretable structure from this heterogenous feature space.

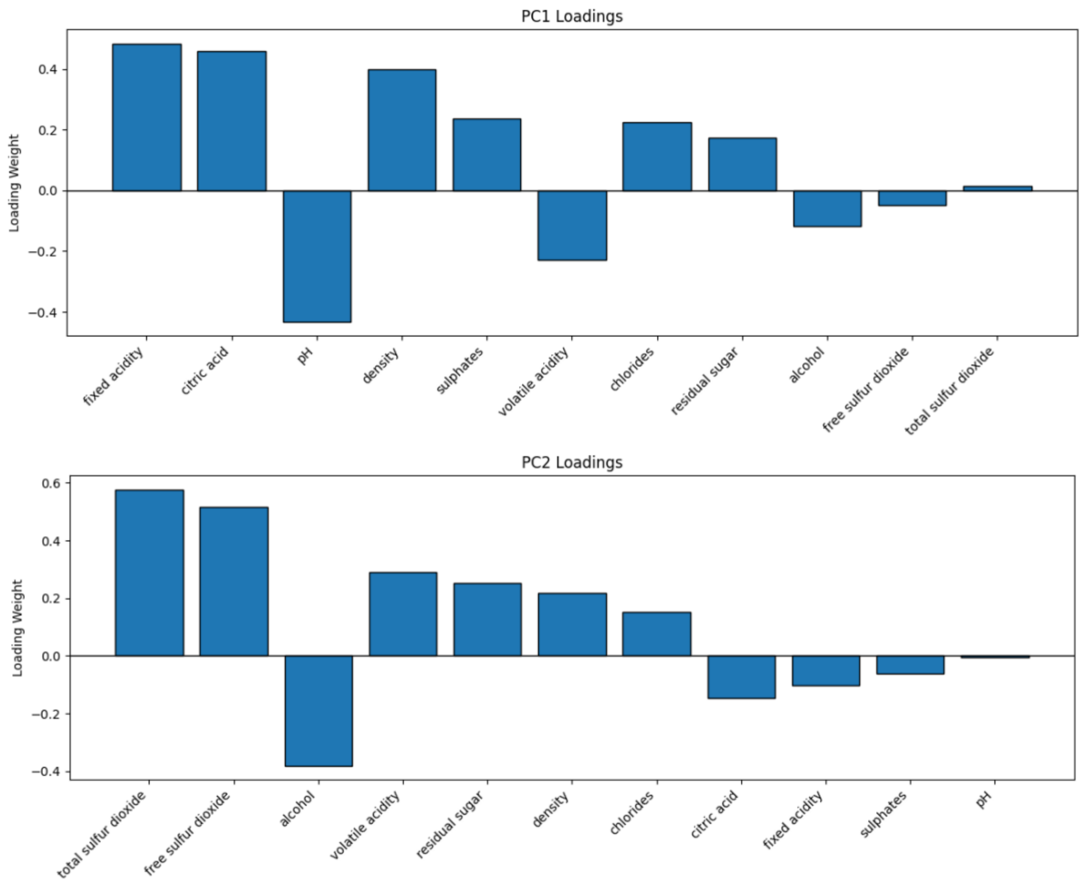

Figure 4.8. Loadings of PC1 and PC2 for the Wine Quality dataset. Each component mixes multiple features with carrying signs, indicating absence of coherent, interpretable latent factor.

This lack of interpretability can be further quantified by examining Pearson correlations between the principal components and the wine quality label. The correlations (PC1: 0.11, PC2: –0.41, PC3: 0.41) indicate that no component exhibits a strong linear relation with quality. Even the most correlated components account for only a small portion of the quality variation, reinforcing that PCA does not recover any meaningful latent axis aligned with wine quality. These results confirm that the dataset's underlying geometry is not compatible with PCA as an interpretability tool.

5. CONCLUSIONS

This study investigated how the interpretability of Principal Component Analysis depends on the underlying structure of high-dimensional representations. Through a comparative analysis of two contrasting data modalities, learned word embeddings and human-engineered feature-based datasets, we demonstrated that PCA does not possess inherent semantic interpretability. Rather, interpretability emerges only when the data itself encodes coherent linear structure.

In the word embedding setting, such structure was present. PCA recovered a clear gender axis, producing meaningful separation between male and female associated words and achieving discriminative performance comparable to the full embedding space. In contrast, the Wine Quality dataset exhibited no dominant linear factors, leading PCA to generate abstract mixtures of features with no interpretable conceptual meaning. Overall, these findings show that PCA can serve as an interpretability tool dependent on the data representation form and on the global linear patterns in the data.

Future work may explore nonlinear dimensionality-reduction methods (e.g., t-SNE, UMAP) or other linear variants designed for improved interpretability (e.g., Sparse PCA, ICA), and broader evaluations across additional embedding models and feature-based datasets. Such directions may further clarify the limitations of PCA and where alternative methods are required for explaining patterns in our data.

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